

A Project Report on

Hostel Complaint Management System

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CERTIFICATE

It is hereby certified that the work which is being presented in the Third Year Project Report entitled "**Hostel Complaint Management System**", in partial fulfillment of the requirements for the award of the Bachelor of Technology in **Computer Engineering** and submitted to the **Department of Computer Engineering of MIT Academy of Engineering, Alandi(D), Pune, Affiliated to Savitribai Phule Pune University (SPPU), Pune**, is an authentic record of work carried out during Academic Year **2025–2026** Semester **VI**.

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Abstract

The Hostel Complaint Management System is a full-stack web application developed using ASP.NET (C#) and MySQL. It is designed to bridge the communication gap between students and hostel administration by providing a structured, digital platform for complaint registration and resolution. Students can submit complaints online and track their status in real-time, while administrators can monitor, manage, and resolve complaints efficiently through a dedicated dashboard. The system employs a role-based access control mechanism, a glassmorphism-inspired UI, and follows the DRY (Don't Repeat Yourself) software design principle to ensure maintainability and scalability.

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1. Introduction

The Hostel Complaint Management System is a web-based application that enables students residing in hostels to register, track, and follow up on complaints related to maintenance, facilities, and services. The system provides hostel administration with a centralized dashboard to receive, process, and resolve complaints in a transparent and efficient manner. Traditionally, complaints were submitted verbally or via paper forms, leading to delays, lost records, and lack of accountability. This system replaces the manual process with a digital workflow that ensures every complaint is logged, assigned a status, and resolved within a traceable lifecycle.

The application is built on the ASP.NET Web Forms framework using C# as the backend language, MySQL as the relational database, and custom CSS3 for the frontend UI. It is hosted on an IIS Express development server and uses ADO.NET for efficient database operations.

2. Background and Motivation

Hostels are an integral part of academic institutions, housing thousands of students who depend on the administration for maintenance and upkeep. The lack of a structured system for lodging complaints often results in delayed responses, unresolved issues, and student dissatisfaction.

This project was motivated by the need to modernize the complaint-handling process at an academic institution. By digitizing the workflow, the system addresses challenges such as:

- Absence of a central repository for complaints
- Inability of students to track complaint status
- Lack of accountability for administrators
- Manual and error-prone record keeping

The development of this system also serves as a practical application of web development concepts including server-side programming, relational database design, session management, and role-based access control.

3. Problem Statement

The existing complaint handling process in hostels is manual and inefficient. Students often face difficulties in getting their issues addressed in a timely manner due to:

- No formal mechanism to record and track complaints
- No transparency between students and hostel management
- Complaints being lost or forgotten without resolution
- No statistical overview available to administrators regarding pending and resolved issues

There is a need for a digital system that provides a structured, transparent, and efficient means for students to raise complaints and for administrators to manage and resolve them with proper documentation.

4. Objectives of the Project

The key objectives of this project are:

1. To provide students with a simple interface to submit and track complaints online.
2. To give hostel administrators a real-time dashboard to monitor complaint statistics and take action.
3. To implement role-based access control distinguishing Student and Admin users.

4. To maintain a complete complaint lifecycle from submission to resolution.
5. To ensure data security through parameterized queries and session-based authentication.
6. To create a visually appealing and responsive UI using modern CSS3 design principles.
7. To follow software engineering best practices such as the DRY principle and separation of concerns.

5. Scope of the Project

The system is designed for use within a single hostel or a group of hostels affiliated with an academic institution. The scope includes:

- User registration and login for students
- Complaint submission with category selection (e.g., Electricity, Plumbing, Cleaning)
- Complaint tracking with real-time status updates
- Admin management of all complaints with the ability to update status and add remarks
- Statistical overview of total, pending, and resolved complaints

The current version is intended for local deployment (IIS Express + XAMPP MySQL). Future extensions may include email notifications, mobile responsiveness enhancements, and multi-hostel support.

6. System Design

6.1 System Architecture

The system follows a three-tier architecture:

- Presentation Layer: ASP.NET Web Forms (.aspx pages) with custom CSS3 and Vanilla JavaScript
- Business Logic Layer: C# code-behind files (.aspx.cs) and the DbHelper utility class
- Data Layer: MySQL database hosted on XAMPP, accessed via ADO.NET

The application uses a Master Page (Site.Master) to ensure consistent navigation, header, and footer across all pages. Role-based routing ensures that students are directed to their workspace and admins are directed to the administrative dashboard upon login.

6.2 Database Design

The database hostel_db consists of two main tables:

Users Table

Field	Data Type	Description
id	INT (PK, AUTO_INCREMENT)	Unique user identifier
username	VARCHAR(50)	Unique login username
password	VARCHAR(100)	User password
full_name	VARCHAR(100)	Full name of the user
email	VARCHAR(100)	Email address
role	ENUM('Student','Admin')	Role for access control
created_at	TIMESTAMP	Account creation time

Complaints Table

Field	Data Type	Description
id	INT (PK, AUTO_INCREMENT)	Unique complaint ID
student_id	INT (FK)	References Users(id)
category	VARCHAR(50)	Complaint category
description	TEXT	Detailed complaint text
status	ENUM('Pending','In Progress','Resolved')	Current complaint status
admin_remarks	TEXT	Admin notes on the complaint
created_at	TIMESTAMP	Submission timestamp
updated_at	TIMESTAMP	Last update timestamp

6.3 Interface Design

The UI is built using a custom CSS file (premium-ui.css) with a Glassmorphism design language — translucent card panels, smooth animations, and gradient backgrounds. Key design features include:

- Dark / Light mode toggle controlled by Vanilla JavaScript using localStorage
- Responsive Master Page layout with consistent navigation
- ASP.NET GridView for tabular complaint display with inline edit functionality
- Dashboard cards showing real-time statistics (Total, Pending, Resolved)

7. Implementation

7.1 Technology Stack

Layer	Technology Used
Frontend	ASP.NET Web Forms, CSS3 (Vanilla), JavaScript (Vanilla)
Backend	C# (ASP.NET)
Database	MySQL via XAMPP
Data Access	ADO.NET with DbHelper.cs
Server	IIS Express (Local Development)
Configuration	XML-based Web.config

7.2 Development Environment

- IDE: Visual Studio 2022
- Database Tool: XAMPP (MySQL + phpMyAdmin)
- Browser: Google Chrome / Microsoft Edge
- OS: Windows 10/11
- Framework: ASP.NET Web Forms (.NET Framework 4.x)

7.3 Module Descriptions

File / Module	Responsibility
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Site.Master	Global layout — header, navigation, footer across all pages
Web.config	Stores DB connection string and app configuration
Login.aspx	Handles authentication and role-based redirection
Register.aspx	Allows students to create accounts
Default.aspx	Landing page; redirects to Login
database.sql	SQL script to create hostel_db, Users and Complaints tables
DbHelper.cs	Centralized ADO.NET utility: ExecuteNonQuery, ExecuteDataTable, ExecuteScalar
Student/Dashboard.aspx	Student summary view — personal complaint statistics
Student/SubmitComplaint.aspx	Form for students to submit new complaints
Student/MyComplaints.aspx	Displays all complaints submitted by the logged-in student
Admin/Dashboard.aspx	Admin control panel — all complaints, inline edit, statistics
premium-ui.css	Entire UI styling — Glassmorphism, animations, themes

7.4 Code Snippets

Key code snippet — DbHelper.cs (Database Utility Class):

```
public static int ExecuteNonQuery(string query, MySqlParameter[] parameters
= null) {
    using (MySqlConnection conn = GetConnection()) {
        using (MySqlCommand cmd = new MySqlCommand(query, conn)) {
            if (parameters != null)
                cmd.Parameters.AddRange(parameters);
            conn.Open();
            return cmd.ExecuteNonQuery();
        }
    }
}
```

Key code snippet — Admin Dashboard (Complaint Update with Parameterized Query):

```
string query = "UPDATE Complaints SET status=@status, admin_remarks=@remarks
WHERE id=@id";
MySqlParameter[] parameters = new MySqlParameter[] {
    new MySqlParameter("@status", newStatus),
    new MySqlParameter("@remarks", remarks),
    new MySqlParameter("@id", complaintId)
};
DbHelper.ExecuteNonQuery(query, parameters);
```

8. Results and Discussion

The Hostel Complaint Management System was successfully implemented and tested. The following results were observed during testing:

8.1 Complaint Lifecycle

8. A student registers and logs in using their credentials.
9. The student submits a complaint (e.g., Category: Plumbing, Description: Leaking pipe).
10. The complaint appears in the student's MyComplaints page with status Pending.
11. The admin logs in, views the complaint on the Admin Dashboard, and updates the status to In Progress with the remark: Plumber assigned.
12. The student logs in again and views the updated status and remark.
13. Upon resolution, the admin sets status to Resolved.

8.2 Key Strengths Observed

- Security: All database queries are parameterized, preventing SQL injection attacks.

- Maintainability: DbHelper.cs and Site.Master reduce code duplication (DRY principle).
- Usability: The Glassmorphism UI with Dark/Light mode was well-received during demonstration.
- Scalability: New complaint categories and admin features can be added with minimal code changes.
- Role Separation: Student and Admin areas are completely isolated with session-based access control.

9. Conclusion and Future Work

The Hostel Complaint Management System successfully addresses the problem of manual, unstructured complaint handling in hostel environments. The system provides a digital, transparent, and efficient platform for both students and administrators, with a complete complaint lifecycle from submission to resolution.

The project demonstrates key software development principles including Separation of Concerns, DRY, role-based access control, and secure database interaction through parameterized queries.

Future Enhancements

- Email/SMS notifications when complaint status is updated
- Priority levels for complaints (High, Medium, Low)
- Mobile-responsive design enhancements for smartphone access
- Multi-hostel support with separate admin accounts per hostel
- Password hashing using bcrypt for improved security
- Analytics dashboard with graphical charts for complaint trends
- Auto-escalation for complaints that remain unresolved beyond a defined deadline

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